FIITJEE

JEE (Main)-2018

ALL INDIA TEST SERIES

PART TEST – III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

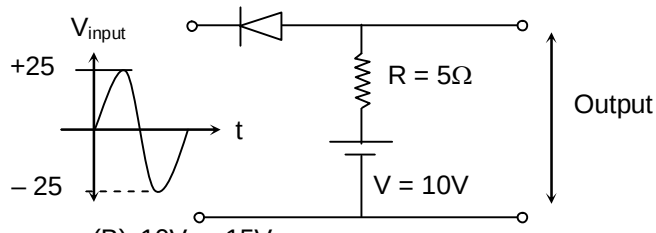
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spherical ball of mass m, radius r is connected to string of length $(\ell >> r)$. This string is then connected to fixed point and this arrangement allowed to oscillate in a medium whose density is ρ and viscosity is η. If amplitude of oscillation becomes half of the initial value after n seconds. Then viscosity of liquid is given by (Assume that resistance force on ball follows stoke's law).

(A) $\frac{m}{\pi rn}$ (B) $\frac{m\ell n\left(2^{1/3}\right)}{\pi rn}$

(C) $\frac{m\ell n8}{\pi rn}$ (D) $\frac{m\ell n2}{\pi rn}$

2. An ideal diode is connected in a circuit with resistance R = 5Ω and V = 10 volt as shown in figure maximum and minimum value of output voltage. When no load applied is (assume diode to be ideal)



(A) 10V, – 25V (B) 10V, – 15V

(C) 25V, – 25V (D) 25V, – 15V

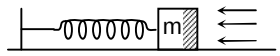
3. A vessel is half filled with a liquid of refractive index μ. The other half of the vessel is filled with an immiscible liquid of refractive index 1.5μ. The apparent depth of the vessel is 50% of the actual depth. Then μ is

(A) 1.4 (B) 1.5

(C) 1.8 (D) 1.67

Space for Rough work

4. A spring block system with mass of block m and spring constant K (all the surfaces of block are perfectly absorbing and smooth) is placed on a smooth horizontal plane as shown in the diagram at rest. A light beam of intensity I is switched on its right side. Find the amplitude of oscillations of the block. (face area of block is A and c is the speed of light in vacuum)



(A) $\frac{I}{Kc}A$ (B) $\frac{2I}{Kc}A$

(C) $\frac{4I}{Kc}A$ (D) Zero

5. What should be the approximate kinetic energy of an electron so that its de-Broglie wavelength is equal to the wavelength of X-ray of maximum energy, produced in an X-ray tube operating at 24800 V? (given that h = 6.6×10^{-34} joule-sec, mass of electron = 9.1×10^{-31} kg)

(A) 600 eV (B) 365 eV

(C) 120 eV (D) 300 eV

6. A 2 kg block moving with 10 m/sec strikes a spring of constant π^2 N/m which is attached to 2 kg block at rest kept on a smooth horizontal floor. The time for which rear moving block remain in contact with spring is

(A) $\sqrt{2}$ sec (B) $\frac{1}{\sqrt{2}}$ sec

(C) 1 sec (D) 1/2 sec

7. Unpolarised light of intensity 32 watt m^{-2} passes through three polarisers such that the transmission axes of the first and second polarises makes an angle 30º with each others the transmission axes of the second and third polarises makes an angle 60º with each other. The intensity of final emerging light will be

(A) 32 watt m^{-2} (B) 3 watt m^{-2}

(C) 8 watt m^{-2} (D) 4 watt m^{-2}.

8. Choose the incorrect statements

(A) An electron beam is used to obtain interference in a simple Young's double-slit experiment. If speed of electron is increased the fringe width decreases.

(B) de-Broglie wavelength of electron is inversely proportional to the speed of electrons.

(C) Two coherent point sources of light having non-zero phase difference are separated by a small distance. On the perpendicular bisector of line segment joining both the point sources, no Constructive interference can be obtained.

(D) For two waves from coherent point sources to interference constructively at a point the magnitude of their phase difference at that point must be $2n\pi$(where n is non-negative integer)

Space for Rough work

9. In radio-receiver, the short wave and medium wave stations transmit signal using same capacitors but ideal coils of different inductors L_s and L_m respectively. Choose the correct option
(A) $L_s > L_m$ (B) $L_s = L_m$
(C) $L_s < L_m$ (D) None.

10. A string of length 2L, obeying Hooke's Law, is stretched so that its extension is L. The speed of the tranverse wave travelling on the string is v. If the string is further stretched so that the extension in the string becomes 4L. The speed of transverse wave travelling on the string will be.
(A) $\sqrt{2}V$ (B) V
(C) $2\sqrt{2}V$ (D) 2V

11. A rigid bar of mass M is supported symmetrically by three wires each of length ℓ. Those at each end are of copper and the middle one is of iron. The ratio of their diameters, if each is to have the same tension, is equal to
(A) $\frac{Y_{copper}}{Y_{iron}}$ (B) $\sqrt{\frac{Y_{iron}}{Y_{copper}}}$
(C) $\frac{Y_{iron}^2}{2Y_{copper}^2}$ (D) $\frac{2}{3}\frac{Y_{iron}}{Y_{copper}}$

12. A close organ pipe of diameter 10 cm has length 42 cm. The air column in pipe vibrates in its second overtone with the maximum amplitude ΔP_o. The pressure amplitude at middle of pipe is
(A) $\frac{\Delta P_o}{2}$ (B) $\frac{\Delta P_o}{\sqrt{2}}$
(C) $\frac{\sqrt{3}\Delta P_o}{2}$ (D) ΔP_o

13. For the same objective, the ratio of the least separation between two points to be distinguished by a microscope using electrons accelerated through 100 V and light of 5500 Å used as the illuminating substance is
(A) 2×10^{-4} (B) 2×10^{-3}
(C) 2×10^{-2} (D) 4×10^{-4}

Space for Rough work

14. M_x and M_y denote the atomic masses of the parent and the daughter nuclei respectively in a radioactive decay. The Q-value for a β-decay is Q_1 and that for a β^+ decay is Q_2. If m_e denotes the mass of an electron, then which of the following statements is correct?

(A) $Q_1 = (M_x - M_y)c^2$ and $Q_2 = (M_x - M_y - 2m_e)c^2$

(B) $Q_1 = (M_x - M_y)c^2$ and $Q_2 = (M_x - M_y)c^2$

(C) $Q_1 = (M_x - M_y - 2m_e)c^2$ and $Q_2 = (M_x - M_y + 2m_e)c^2$

(D) $Q_1 = (M_x - M_y + 2m_e)c^2$ and $Q_2 = (M_x - M_y + 2m_e)c^2$

15. Tritium is an isotope of hydrogen whose nucleus Triton contains 2 neutrons and 1 proton. Free neutrons decay into $p + \bar{e} + \bar{\nu}$ If one of the neutrons in Triton decays, it would transform into a He^3 nucleus. This does not happen because

(A) Triton energy is less than that of a He^3 nucleus.
(B) The electron created in the beta decay process cannot remain in the nucleus.
(C) Both the neutrons in triton have to decay simultaneously resulting in a nucleus with 3 protons, which is not a He^3 nucleus.
(D) Because free neutrons decay due to external perturbations which is absent in a triton nucleus.

16. A tuning fork A is being tested using an accurate oscillator. It is found that they produce 2 beats per second when the oscillator reads 514 Hz and 6 beats per second when it reads 510 Hz. The actual frequency of the fork in Hz is

(A) 512 Hz (B) 504 Hz
(C) 516 Hz (D) 510 Hz

17. The objective of telescope has diameter 12 cm. The distance at which two small green object placed 30 cm apart can be barely resolved by telescope, assuming the resolution to be limited by diffraction by objective only ($\lambda = 5.4 \times 10^{-5}$ cm)

(A) 54.6 km (B) 57.0 km
(C) 546 km (D) 52.4 km

Space for Rough work

18. Two cars A and B are moving towards each other with same speed 25 m/s. Wind is blowing with speed 5 m/s in the direction of motion of car A. The car A blows horn of frequency 300 Hz and the sound is reflected from car B. The wavelength of the reflected sound received by the driver of car A is (velocity of sound in air = 330 m/s)

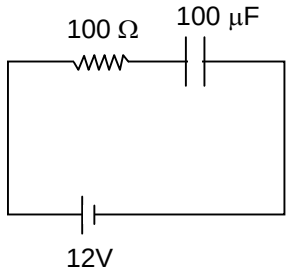
(A) $\frac{25}{36}$m (B) $\frac{31}{36}$m

(C) $\frac{35}{36}$m (D) $\frac{37}{36}$m

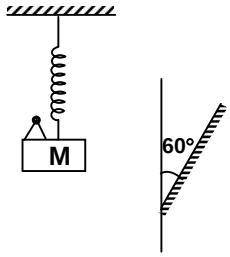
19. The value of displacement current at t = 1 time constant is

(A) $\frac{120}{e}$mA (B) $\frac{120}{e^2}$mA

(C) 120 mA (D) None of these



20. An insect of negligible mass is sitting on a block of mass M, tied with a spring of force constant k. The block performs simple harmonic motion with amplitude A in front of a plane mirror placed as shown in the figure. The maximum speed of insect relative to its image will be



(A) $A\sqrt{\frac{k}{m}}$ (B) $\frac{A\sqrt{3}}{2}\sqrt{\frac{k}{m}}$

(C) $A\sqrt{3}\sqrt{\frac{k}{m}}$ (D) $A\sqrt{\frac{m}{k}}$

21. Mass of a sphere of radius r = 10.00 cm is 30.000 g
If measured values are expressed upto the correct significant figures then maximum fractional error in calculation of density of the sphere is

(A) 3×10^{-5} (B) 3.3×10^{-3}

(C) 3.033×10^{-3} (D) 3.03×10^{-5}

Space for Rough work

22. Maximum pressure difference between inner and outer surface of a thin walled circular tube of radius r and thickness $\Delta r\left(\Delta r << r\right)$, so that tube won't break is (Breaking stress of the material is σ_0)

(A) $\frac{\sigma_0 r}{\Delta r}$ (B) $\frac{\Delta r \sigma_0}{r}$

(C) $\frac{2\sigma_0 r}{\Delta r}$ (D) σ_0

23. $\frac{7}{8}$th of nuclei have been disintegrated in 2 minutes after fresh sample was prepared. The half life of the sample is

(A) 30 seconds (B) 1 minute

(C) 40 seconds (D) 20 seconds

24. An astronomical telescope has angular magnification m = 7 for distant object. Distance between objective and eyepiece is 40 cm. Final image is formed at infinitely. Focal length of objective and eyepiece respectively are:

(A) 35 cm, 5 cm (B) 28 cm, 14 cm

(C) 5 cm, 35 cm (D) 70 cm, 10 cm

25. The length of an elastic chord is ℓ_1 when tension in it is T_1 and ℓ_2 when tension is T_2. The natural length of wire is

(A) $\frac{\ell_1 T_1 + \ell_2 T_2}{T_1 + T_2}$ (B) $\frac{\ell_2 T_1 - \ell_1 T_2}{T_1 - T_2}$

(C) $\frac{\ell_2 T_1 + \ell_1 T_2}{T_1 + T_2}$ (D) $\frac{\ell_1 T_1 - \ell_2 T_2}{T_1 - T_2}$

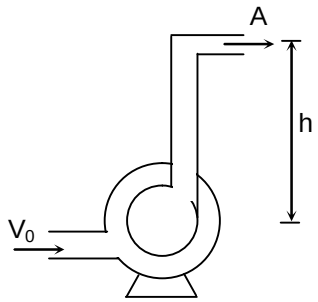
26. The angular speed of rotation of earth about its axis when the weight of a man standing on equator will become half of its weight at the north pole (assume earth to be a perfect sphere) R-6400 km.

(A) 275×10^{-4} rad sec^{-1} (B) 9.75×10^{-4} rad sec^{-1}

(C) 8.75×10^{-4} rad sec^{-1} (D) 8.0×10^{-4} rad sec^{-1}

Space for Rough work

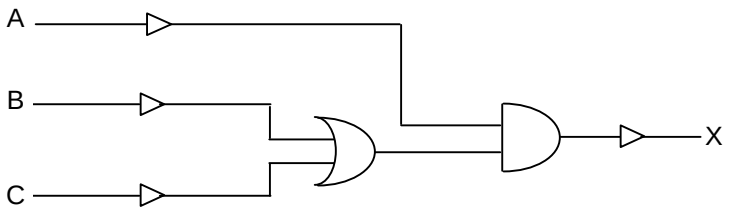
27. In a certain pump whose rated power is P. Ideal fluid of density ρ enters with speed V_0 which lifts water upto height h through a pipe of uniform cross section area without discontinuity. Volumetric flow rate of the liquid is Q. Speed of fluid when it comes out of the pipe at point A is {Assume that pump works at 100% efficiency and flow is streamline}



(A) $\sqrt{\frac{2P}{Q\rho}+V_0^2-2gh}$ (B) $\sqrt{\frac{2P}{Q\rho}+V_0^2}$

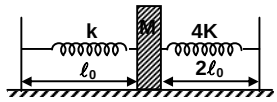
(C) $\sqrt{\frac{2P}{Q\rho}+V_0^2+2gh}$ (D) $\sqrt{\frac{2P}{Q\rho}-V_0^2+2gh}$

28. In above diagram × = ?
(A) A + BC
(B) AB + C
(C) AC + B
(D) None of above



29. An object is floating in water with fraction of its volume submerged in water filled in beaker which is at rest with respect to ground. Now this beaker is accelerated with constant acceleration downwards (a < g). Then,
(A) Fraction of volume submerged will not change.
(B) Buoyant force on the block will remain same.
(C) Pressure at any point in the water will be more than earlier.
(D) Fraction of volume submerged will change.

30. A block of mass M is kept in gravity free space and touches the two springs as shown in the figure. Initially springs are in its natural length. Now, the block is shifted ($\ell_0/2$) from the given position in such a way it compresses a spring and released. The time-period of oscillation of mass will be



(A) $\frac{\pi}{2}\sqrt{\frac{M}{K}}$ (B) $2\pi\sqrt{\frac{M}{5K}}$

(C) $\frac{3\pi}{2}\sqrt{\frac{M}{K}}$ (D) $\pi\sqrt{\frac{M}{2K}}$

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. The pair of compounds having metals in their highest oxidation state
(A) MnO_2, $FeCl_3$ (B) $[MnO_4]^-$, CrO_2Cl_2
(C) $[Fe(CN)_6]^{3-}$, $[Co(CN)_3]$ (D) $[NiCl_4]^{2-}$, $[CoCl_4]^-$

32. The density of solid Ar (Ar = 40 g/mole) is 1.68 g/ml at 40 K. If the argon atom is assumed to be a sphere of radius 1.50×10^{-8} cm, then % of solid Ar is apparently empty space?
(A) 35.64 (B) 64.36
(C) 74 (D) None these

33. Resistance of a decimolar solution between two electrodes is 0.02 meter apart and 0.0004 m^2 in area was found to be 50 ohm. Specific conductance will be
(A) 0.1 Scm^2 (B) 1 Scm^2
(C) 10 Scm^2 (D) 4×10^{-4} Scm^2

34. Poling process is used for
(A) the removal of Cu_2O from Cu (B) the removal of Al_2O_3 from Al
(C) the removal of Fe_2O_3 from Fe (D) All of these

35. A solution of $Na_2S_2O_3$ is standarised idometrically against 0.167 gram of $KBrO_3$. This process requires 50 mL of the $Na_2S_2O_3$ solution. What is the normality of the $Na_2S_2O_3$?
(A) 0.2 N (B) 0.12 N
(C) 0.72 N (D) 0.02 N

36. A 0.10 M solution of a mono protic acid (d = 1.01 g/cm^3) is 5% dissociated. What is the freezing point of the solution. The molar mass of the acid is 300 and $K_f(H_2O) = 1.86^oC/m$
(A) -0.189^oC (B) -0.194^oC
(C) -0.199^oC (D) None

Space for Rough work

37. In a spinel structure of a mixed oxide, oxide ion are arranged in FCC packing whereas $\frac{1}{8}$th of tetrahedral voids are occupied by A^{2+} ion and $\frac{1}{2}$ of octahedral voids are occupied by B^{3+} cations. The general formula of the compound having structure is
(A) $A_2B_2O_4$ (B) AB_2O_4
(C) $A_2B_4O_2$ (D) $A_4B_2O_2$

38. In the purification of impure Ni by Mond's process, metal is purified by
(A) Electrolytic reduction (B) Vapour phase thermal decomposition
(C) Thermite reduction (D) Carbon reduction

39. $\underset{(\text{Black})}{X} \xrightarrow{\text{Dil. } H_2SO_4} \underset{(\text{Gas})}{Y} \xrightarrow{\text{Dil. } HNO_3} \text{Colloidal sulphur}$
Identify X?
(A) CuS (B) FeS
(C) As_2S_3 (D) CdS

40. Fixed mass of an ideal gas collected in a 24.63 litre sealed rigid vessel at 1 atm is heated from – 73°C to 27°C. Calculate change in Gibbs free energy if entropy of gas is a function of temperature as $S = 2 + 10^{-2}\,T$ (J/K) (1 atm = 0.1 kJ)
(A) 1231.5 J (B) 1281.5 J
(C) 785.1 J (D) None

41. CsCl has bcc structure with Cs^+ at the centre and Cl^- ion at each corner. If r_{Cs^+} is 1.69 $\mathring{A}$ and r_{Cl^-} is 1.81 $\mathring{A}$. What is the edge length of the cube?
(A) 3.5 $\mathring{A}$ (B) 3.80 $\mathring{A}$
(C) 4.04 $\mathring{A}$ (D) 4.5 $\mathring{A}$

42. The cell $Pt\,|\,H_2\,(g, 0.1\,bar)\,|\,H^+\,(aq), pH = x\,||\,Cl^-\,(1M)\,|\,Hg_2Cl_2\,|\,Hg\,|\,Pt$ has emf of 0.5755 V at 25°C. The SOP of Calomel electrode is – 0.28 V, then pH of the solution will be
(A) 11 (B) 4.5
(C) 5.5 (D) None

Space for Rough work

43. For a reversible adiabatic ideal gas expansion $\frac{dp}{p}$ is equal to

(A) $\gamma\frac{dV}{v}$ (B) $-\gamma\frac{dv}{v}$

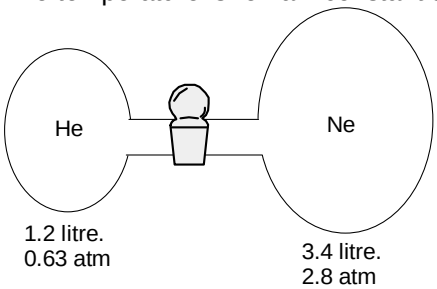
(C) $\left(\frac{\gamma}{\gamma-1}\right)\frac{dV}{V}$ (D) $\frac{dV}{V}$

44. A freshly prepared $Fe(OH)_3$ precipitate is peptized by adding $FeCl_3$ solution. The charge on the colloidal particle is due to preferential adsorption of

(A) Cl^- (B) Fe^{3+}

(C) OH^- (D) None of these

45. Consider the following apparatus. Calculate the partial pressure of He after opening the valve. The temperature is remain constant at 16^oC



(A) 0.164 atm (B) 1.64 atm

(C) 0.328 atm (D) 1 atm

46. On adding KI to a metal salt solution, no precipitate was observed but the salt solution gives yellow precipitate with K_2CrO_4 in the presence CH_3COOH. Then the salt is

(A) $Sr(NO_3)_2$ (B) $Pb(CH_3COO)_2$

(C) $AgNO_3$ (D) $BaCl_2$

47. Which of the following is true in respect of chemical adsorption?

(A) $\Delta H < O\ \ \Delta S > O\ \ \Delta G > O$ (B) $\Delta H < O\ \ \Delta S < O\ \ \Delta G < O$

(C) $\Delta H > O\ \ \Delta S > O\ \ \Delta G < O$ (D) $\Delta H > O\ \ \Delta S < O\ \ \Delta G > O$

48. For the reaction $N_2H_4 + IO_3^- + 2H^+ + Cl^- \longrightarrow ICl + N_2 + 3H_2O$

The eq. mass of N_2H_4 and KIO_3 is respectively are

(A) 8 and 53.5 (B) 16 and 53.5

(C) 8 and 35.6 (D) 8 and 87

Space for Rough work

49. Which of the following statement is false?
(A) Optical isomerism is observed in $[M_{abcd}]^{n\pm}$ tetrahedral complexes.
(B) Geometrical isomerism does not exists while optical isomerism exists in the complex $\left[Fe(C_2O_4)_3\right]^{3-}$.
(C) Both *cis* and *trans* forms are optically inactive in $\left[PtCl_2(NH_3)_4\right]^{2+}$ complex ion.
(D) $\left[Pt(en)_2\right]^{2+}$ shows geometrical isomerism as well as optical.

50. Consider the following reaction:
$H_2O(\ell) \longrightarrow H_2O(g) \quad \Delta H_1 = 44$ kJ
$2CH_3OH(\ell) + 3O_2 \longrightarrow 4H_2O(\ell) + 2CO_2(g) \quad \Delta H_2 = -1453$ kJ
What is the value of ΔH for second reaction if water vapour instead of liquid water is formed as product?
(A) - 1409 kJ (B) - 1629 kJ
(C) - 1277 kJ (D) None of these

51. Smog is essentially caused by the presence of
(A) O_3 and N_2 (B) O_2 and N_2
(C) Oxides of sulphur and N_2 (D) O_2 and O_3

52. Which of the following is not tranquilizer?
(A) Luminal (B) Seconal
(C) Reserpine (D) Piperazine

53. Which of the following will not be oxidised by O_3?
(A) KI (B) $FeSO_4$
(C) $KMnO_4$ (D) K_2MnO_4

54. Which of the following process is used in the extractive metallurgy of Mg?
(A) Fused salt electrolysis (B) Self – reduction
(C) Aqueous solution electrolysis (D) Thermite reduction

55. Which of the following compound is not an antacid?
(A) Aluminium hydroxide (B) Cimetidine
(C) Phenelzine (D) Ranitidine

56. The gas leaked from a storage tank of the union carbide plat in Bhopal gas tragedy was
(A) Ammonia (B) Phosgene
(C) Methyl isocynate (D) Methyl amine

Space for Rough work

57. 36 mL of pure water takes 100 sec to evaporate from a vessel when a heater of 806 watt is used. The $\Delta H_{vaporisation}$ of H_2O is: (density of water = 1 g/cc)

(A) 40.3 kJ/mole (B) 43.2 kJ/mole
(C) 4.03 kJ/mole (D) none

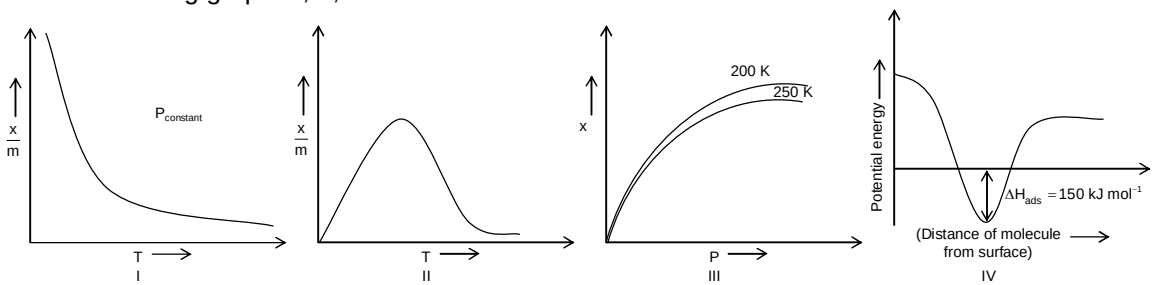
58. 108 g fairly concentrated solution of $AgNO_3$ is electrolysed by using 0.1 F charge. The mass of resulting solution is

(A) 94 g (B) 11.6 g
(C) 96.4 g (D) None

59. The geometry, hybridization and magnetic moment of $[Ni(CO)_4]$ are……..,….and …..respectively

(A) tetrahedral, sp^3, zero (B) square planar, sp^3, $\sqrt{3}$
(C) tetrahedral, sp^3, $2\sqrt{2}$ (D) square planar, dsp^2, zero

60. X = amount of gas adsorbed
P = Pressure
T = Temperature
L = Physisorption
M = Chemisorption
For the following graphs I, II, III and IV



Chose correct combination:

(A) I → L, II → M (B) I → L, III → M
(C) IV → M, II → L (D) IV → M, III → M

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. If $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$, $\left(\vec{a} \wedge \vec{b}\right) = \frac{\pi}{2}$, and $\vec{a} \cdot \vec{c} = 4$, then

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = |\vec{a}|$ (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{a}|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{a}|^2$

62. The value of m for which the straight line $2x - 3y + 4 = 0 = x - 4y + 4z + 5$ is parallel to the plane $2x - y + mz - 2 = 0$ is

(A) 3 (B) 2

(C) $\frac{-16}{5}$ (D) –4

63. A line L passing through the point P(0, 1, –1), and is perpendicular to both the lines $\frac{x-2}{2} = \frac{y-4}{1} = \frac{z+2}{4}$ and $\frac{x+2}{3} = \frac{y+4}{2} = \frac{z-1}{-2}$. If the position vector of point Q on L is (a, b, c) such that $(PQ)^2 = 357$, then $a + 2b + 3c$ is equal to

(A) 26 (B) 24

(C) –24 (D) 7

64. Let O be the interior point of ΔABC such that $3\overrightarrow{OA} + 4\overrightarrow{OB} + 6\overrightarrow{OC} = 0$, then $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOB}$ is equal to

(A) $\frac{13}{7}$ (B) $\frac{13}{6}$

(C) $\frac{8}{7}$ (D) $\frac{7}{13}$

Space for Rough work

65. If $\vec{a} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} - 3\hat{k}$ then the vector $\left(\hat{a} \times \hat{b}\right) \times \left(\hat{a} + \hat{b}\right)$ is collinear to the vector

(A) $4\hat{i} + \hat{j} + 4\hat{k}$ (B) $\hat{i} + 5\hat{j} + 6\hat{k}$

(C) $5\hat{i} + 4\hat{j} + 5\hat{k}$ (D) $5\hat{i} + 5\hat{k}$

66. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero non-coplanar vectors and $\vec{x} = \vec{a} + 2\vec{b} - \vec{c}$, $\vec{y} = \vec{a} - 2\vec{b} - 2\vec{c}$ and $\vec{z} = \vec{a} - 4\vec{b} + 2\vec{c}$ are three vectors such that volume of the parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ and $\vec{x}, \vec{y}, \vec{z}$ as their coterminous edges are v_1 and v_2 respectively. Then $\frac{v_2}{v_1}$ is

(A) 10 (B) 15

(C) 18 (D) none of these

67. If $\sum_{k=0}^{100} \left(\frac{k}{k+1}\right)^{100} C_k = \frac{x\left(2^{100}\right) + y}{z}$ where x, y, z $\in$ N then least value of xy^2z is equal to

(A) 9999 (B) 1010

(C) 9090 (D) none of these

68. A two digit positive number is selected at random then the probability that its tens digit is at least four more than its unit digit, is

(A) $\frac{1}{6}$ (B) $\frac{7}{30}$

(C) $\frac{14}{45}$ (D) none of these

69. Consider the system of equations $\alpha x - y + z = \alpha$; $x - \alpha y + z = 1$; $x - y + \alpha z = 1$. If L, M, and N denotes the number of integral values of α in interval [–5, 7] for which the system of equation has unique solution, no solution and infinite solution respectively, then the value of $L^2 + M^2 + N^2$ is

(A) 125 (B) 13

(C) 15 (D) 121

70. If α, β, γ be the roots of the equation $3x^3 - 5x^2 + 6x - 2 = 0$ and A(α, β, γ), B(β, γ, α), C(γ, α, β) are the vertices of the triangle, then centroid of ΔABC lies on the line

(A) x = 2y = 3z (B) x = y = z

(C) x = –2y = z (D) –x = y = –z

Space for Rough work

71. The coefficient of x^{20} in the polynomial $(1+x)^{22} + x(1+x)^{21} + x^2(1+x)^{20} + ... + x^{20}(1+x)^2$ is
(A) $^{22}C_{19}$ (B) $^{23}C_3$
(C) $^{22}C_2$ (D) $^{23}C_{19}$

72. Number of complex numbers z such that $|z| < \frac{1}{3}$ and $\sum_{r=1}^{n} a_r z^r = 1$ (where $|a_r| < 2$) is
(A) 1 (B) 2
(C) n (D) 0

73. Let $A_k = \cos\frac{k\pi}{20} + i\sin\frac{k\pi}{20}$ where $\sqrt{-1} = i$, then $\frac{\sum_{k=0}^{2018} |A_{k+1} - A_k|}{\sum_{k=1}^{673} |A_{2k+1} - A_{2k+2}|}$ is equal to
(A) 3 (B) $\frac{2018}{673}$
(C) $\frac{2019}{674}$ (D) none of these

74. The value of $\begin{vmatrix} (a-p)^{-2} & (a-p)^{-1} & p^{-1} \\ (a-q)^{-2} & (a-q)^{-1} & q^{-1} \\ (a-r)^{-2} & (a-r)^{-1} & r^{-1} \end{vmatrix}$ is
(A) $\frac{a^2\prod(p-q)}{pqr\prod(a-p)^2}$ (B) $\frac{-a^2\prod(p-q)}{prq\prod(a-p)^2}$
(C) $\frac{b^2\prod(p-q)}{pqr\prod(a-p)^2}$ (D) $\frac{c^2\prod(p-q)}{pqr\prod(a-p)^2}$

Space for Rough work

75. If a_r, b_r, c_r (r = 1, 2, 3) are non-negative real numbers and $\sum_{r=1}^{3}(a_r + b_r + c_r) = A$, then maximum value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ is

(A) A^3 (B) $\frac{A^3}{8}$

(C) $\frac{A^3}{27}$ (D) $\frac{A^2}{9}$

76. If a point z_1 satisfying $|z + 8| \le 2$ and z_2 satisfying $|z + 3| + |z - 3| \le 10$ then the range of $|z_1 - z_2|$ is

(A) [10, 15] (B) [1, 15]

(C) [2, 10] (D) none of these

77. Area of the triangle whose vertices are A(–3 – 2i), B(2 – 3i) and C(–1 –5i), is

(A) $\frac{\sqrt{7}}{4}$ (B) $\frac{\sqrt{14}}{2}$

(C) $\frac{\sqrt{13}}{2}$ (D) $2\sqrt{14}$

78. If k_1 is the number of arrangements of the letters of the word "EQUATION" in which all vowels are together and k_2 is the number of arrangements of the letters of the word "EQUATION" in which vowels are in alphabetical order then $\frac{7k_1}{k_2}$ is

(A) 120 (B) $4! \times 5!$

(C) 60 (D) $\frac{8!}{5!}$

79. 100 identical objects are distributed among 10 persons in which 1st person gets at least one object, 2nd person gets at least 2 objects, 3rd person gets at least 3 objects, ..., 10th person gets at least 10 objects, then total number of ways of distribution is

(A) ${}^{55}C_{10}$ (B) ${}^{55}C_9$

(C) ${}^{54}C_9$ (D) ${}^{54}C_{44}$

Space for Rough work

80. A palindrome is a word, number, phrase or sequence of words that reads the same backwards as forwards e.g. "SOLOS". The number of palindromes that can be formed using the letters AABBBBCCCDDDD are

(A) $\frac{6!}{2!}$ (B) $\frac{6!}{2! \times 2!}$

(C) $\frac{7!}{2! \times 2!}$ (D) none of these

81. Two cards are missing from a pack of 52 playing cards. One card is selected at random then the probability that it is a king, is

(A) $\frac{3}{26}$ (B) $\frac{1}{13}$

(C) $\frac{10}{13}$ (D) $\frac{4}{13}$

82. All the words formed using the letters of the word "ALGEBRA" taken all at a time and one of the word is selected from it, then the probability that the word contains the word "AGE", is

(A) $\frac{^7C_2}{5!}$ (B) $\frac{5!}{7!}$

(C) $\frac{1}{^7P_2}$ (D) $\frac{1}{^7C_2}$

83. If $I_n = \int_0^{\pi/4} \tan^n x dx$ and $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}, \frac{1}{I_5 + I_7}$ form an A.P., then the common difference of the A.P. is

(A) 1 (B) 2

(C) 3 (D) none of these

84. The sum of the series $\frac{8}{5} + \frac{16}{65} + \frac{24}{325} + \frac{32}{1025}$ ..... upto 20 terms is

(A) $\frac{1280}{841}$ (B) $\frac{1680}{841}$

(C) $\frac{1880}{841}$ (D) none of these

Space for Rough work

85. If x^2, A_1, A_2, y^2 are in arithmetic progression, x^2, G_1, G_2, y^2 are in geometric progression and x^2, H_1, H_2, y^2 are in harmonic progression, then the value of $(G_1G_2)(H_1 + H_2) - H_1H_2(A_1 + A_2)$ is
(A) 0 (B) 1
(C) 2 (D) none of these

86. Let A and B are two events such that P(A) = 0.4, P(B) = 0.6 and $P\left(\frac{B}{A}\right) = 0.5$. Then $P\left(\frac{\bar{A}}{\bar{B}}\right)$ equals to
(A) $\frac{1}{4}$ (B) $\frac{2}{3}$
(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

87. If $f(x) = x^2 + \alpha x + \beta$ (where $\alpha, \beta \in R$) and $f(f(x)) = 0$ has two roots 1 and 2, then f(0) is equal to
(A) 1 (B) –2
(C) $\frac{-3}{2}$ (D) –1

88. A bag contains 5 balls of unknown colours. A ball is drawn at random from it and is found to be white. The probability that bag contains only white balls is
(A) $\frac{3}{5}$ (B) $\frac{1}{5}$
(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

89. The arithmetic mean of a set of observation is $\bar{x}$. If each observation is divided by α and then increased by 10, then mean of the new series is
(A) $\frac{\bar{x}}{\alpha}$ (B) $\frac{\bar{x}+10}{\alpha}$
(C) $\frac{\bar{x}+10\alpha}{\alpha}$ (D) $\alpha\bar{x}+10$

90. The coefficient of x^{203} in the expression $(x - 1)(x^2 - 2)(x^3 - 3) ..... (x^{20} - 20)$ is
(A) 11 (B) 12
(C) 13 (D) 15

Space for Rough work